DO NOW — RECALL

Software development: practical testing

Name: _____ Class: _____ Date: _____

1. What is a prototype?

2. Why might a tester run tests at each sprint?

3. What does AND require for a compound condition to be true?

4. For valid ages 12–16 inclusive, which four values test either side of both boundaries?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

An early model that users evaluate and refine.

2. Model answer

To catch regressions while new features are introduced.

3. Model answer

Both conditions must be true.

4. Model answer

11, 12, 16 and 17.

Vocabulary — know these terms

normal test — Typical valid input.

boundary test — Input at or near a valid limit.

erroneous test — Input outside permitted requirements.

expected result — Predicted correct output for a test.

regression — New change breaks previously working behaviour.

NEW LEARNING

Read, annotate and explain

State the valid range

The specification permits whole-number marks from 0 through 100 inclusive.

Choose targeted tests

Normal value 50; valid boundaries 0 and 100; invalid just beyond boundaries −1 and 101.

Write expected results

A useful test log records input, expected outcome and actual outcome; a mismatch is evidence of a fault.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

Design tests for a password length from 8 through 20 characters inclusive.

Predict first

Test just below and at the lower limit: 7 rejected, 8 accepted.

Model and explain

Also test a normal 14, upper boundary 20, and 21 rejected.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

A whole-number temperature must be 10–40 inclusive. Give a normal, a lower-boundary, an upper-boundary and an erroneous test.

Question 2

Write C# to validate a score from 1 to 9 inclusive and give four expected-result test pairs.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

A developer tests values 20,30,40 for a permitted range 20–40. Explain why this is insufficient and propose a stronger suite.

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

Testing only typical valid inputs proves that validation rejects out-of-range values.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.